A Lehmann–Romano Proof of Admissibility for Closed Convex Acceptance Regions
Updated V2

1 Lehmann–Romano supported formulation

Lehmann and Romano's Section 6.7 gives a sufficiency result. In the formulation used below, a closed convex acceptance region, together with the ray condition in Theorem 6.7.1, implies d-admissibility. If, in addition, the test has size exactly α at a finite point in the closure of the null parameter space, then Corollary 6.7.1 gives α-admissibility.

Thus the conclusion is

$$A_0 \text{ closed and convex} \quad \Longrightarrow \quad d\text{-admissibility},$$

and, with exact size attained at a finite null point,

$$A_0 \text{ closed and convex} \quad \Longrightarrow \quad \alpha\text{-admissibility}.$$

The geometric containment conclusion in Theorem 6.7.1 is stated for ordinary acceptance regions, hence for nonrandomized competitors. The admissibility conclusion below is obtained by invoking the Lehmann–Romano corollary directly. This is important because Lehmann and Romano note that Theorem 6.7.1 and Corollary 6.7.1 extend to randomized competitors, while the maintained test ϕ_0 itself must be nonrandomized.

2 The theorem and proof

Theorem 1 (Closed convex acceptance regions imply admissibility)**.** *Let* $0 < \alpha < 1$. *Let*

$$X \sim N(\mu, \Sigma),$$

where Σ *is known and positive definite and* $\mu \in \mathbf{R}^k$. *Consider testing*

$$H_0 : \mu \in \Lambda_0 = \{\mu : \mu \leq 0\} \quad \textit{versus} \quad H_1 : \mu \in \Lambda_1 = \mathbf{R}^k \setminus \Lambda_0.$$

Let ϕ_0 *be a nonrandomized test with acceptance region*

$$A_0 = \{x : \phi_0(x) = 0\}.$$

Assume:

(i) A_0 *is closed and convex;*

(ii) ϕ_0 has level α:

$$\sup_{\mu \in \Lambda_0} E_\mu[\phi_0(X)] \le \alpha.$$

Then ϕ_0 is d-admissible for testing H_0 against H_1. If, in addition, there exists a finite point $\mu_0 \in \Lambda_0$ such that

$$E_{\mu_0}[\phi_0(X)] = \alpha,$$

then ϕ_0 is α-admissible. Both admissibility conclusions are with respect to possibly randomized competitors.

Proof. We use the following consequence of Theorem 6.7.1 and Corollary 6.7.1 of Lehmann and Romano.

Proposition 1 (Lehmann–Romano)**.** *Consider an exponential family with density*

$$p_\theta(t) = C(\theta) \exp\{\theta' t\}$$

with respect to a σ-finite measure ν, with natural parameter space Ω. Let Ω_H and Ω' be disjoint nonempty subsets of Ω. Let ϕ_0 be a nonrandomized test of $H : \theta \in \Omega_H$, based on T, with acceptance region $A_0 \subseteq \mathbf{R}^s$. Suppose A_0 is closed and convex and that the following ray condition holds: whenever

$$A_0 \cap \{t : a't > c\} = \varnothing,$$

there exist $\theta^* \in \Omega$ and $\lambda_n \to \infty$ such that

$$\theta^* + \lambda_n a \in \Omega'.$$

Then the test with acceptance region A_0 is d-admissible. If the test has size α and there exists a finite point $\theta_0 \in \overline{\Omega_H}$ such that

$$E_{\theta_0}[\phi_0(X)] = \alpha,$$

then the test is also α-admissible. These admissibility conclusions allow randomized competing tests.

We now verify the assumptions of this result. Write the normal density in canonical exponential-family form. Put

$$\theta = \Sigma^{-1}\mu, \qquad \mu = \Sigma\theta.$$

With respect to the measure

$$d\nu(x) = (2\pi)^{-k/2} |\Sigma|^{-1/2} \exp\left\{ -\frac{1}{2} x' \Sigma^{-1} x \right\} dx,$$

the density of $X \sim N(\Sigma\theta, \Sigma)$ is

$$p_\theta(x) = \exp\left\{ \theta' x - \frac{1}{2} \theta' \Sigma \theta \right\}.$$

Thus $T(x) = x$, $\Omega = \mathbf{R}^k$, and

$$\Omega_H = \{\theta : \Sigma\theta \le 0\}, \qquad \Omega_K = \{\theta : \Sigma\theta \not\le 0\}.$$

We apply the Lehmann–Romano result with $\Omega' = \Omega_K$.

It remains to verify the ray condition. Let $a \in \mathbf{R}^k$ and $c \in \mathbf{R}$ satisfy

$$A_0 \cap \{x : a'x > c\} = \varnothing.$$

If $a = 0$, choose any $\theta^* \in \Omega_K$ and the condition is immediate. Hence suppose $a \neq 0$. Equivalently, $A_0 \subseteq \{x : a'x \leq c\}$. We claim that

$$\Sigma a \not\leq 0.$$

Suppose instead that $\Sigma a \leq 0$. For $\lambda > 0$, define

$$\theta_\lambda = \lambda a, \qquad \mu_\lambda = \Sigma\theta_\lambda = \lambda\Sigma a.$$

Then $\mu_\lambda \leq 0$, so $\theta_\lambda \in \Omega_H$. Under P_{θ_λ},

$$a'X \sim N\left(\lambda a'\Sigma a,\ a'\Sigma a\right).$$

Since Σ is positive definite and $a \neq 0$, $a'\Sigma a > 0$. Hence

$$P_{\theta_\lambda}(a'X > c) \to 1.$$

Because $A_0 \subseteq \{x : a'x \leq c\}$, we have $\{x : a'x > c\} \subseteq A_0^c$. Since ϕ_0 is nonrandomized,

$$E_{\theta_\lambda}[\phi_0(X)] \geq P_{\theta_\lambda}(a'X > c) \to 1.$$

This contradicts the level condition because every $\theta_\lambda \in \Omega_H$ and $\alpha < 1$. Therefore $\Sigma a \not\leq 0$.

It follows that $\lambda\Sigma a \not\leq 0$ for every $\lambda > 0$. Hence

$$\lambda a \in \Omega_K \qquad \text{for every } \lambda > 0.$$

The ray condition holds with $\theta^* = 0$ and any sequence $\lambda_n \to \infty$.

The Lehmann–Romano proposition therefore gives d-admissibility of ϕ_0 against possibly randomized competitors.

Finally suppose there exists finite $\mu_0 \in \Lambda_0$ such that $E_{\mu_0}[\phi_0(X)] = \alpha$. Let $\theta_0 = \Sigma^{-1}\mu_0$. Then $\theta_0 \in \Omega_H \subseteq \overline{\Omega_H}$ and

$$E_{\theta_0}[\phi_0(X)] = \alpha.$$

The second part of the Lehmann–Romano corollary gives α-admissibility, again against possibly randomized competitors. □

3 Why exact size is needed for α-admissibility

The statement "closed and convex acceptance region implies α-admissibility" is false if one only assumes

$$\sup_{\mu \leq 0} E_\mu[\phi(X)] \leq \alpha.$$

For example, the zero test $\phi(x) \equiv 0$ has acceptance region $\mathbf{R}^k$, which is closed and convex, and it satisfies the size constraint for every $0 < \alpha < 1$. But it is not α-admissible. The test

$$\psi(x) = 1\left\{x_1 > \sqrt{\Sigma_{11}}\Phi^{-1}(1-\alpha)\right\}$$

has level α for $H_0 : \mu \leq 0$ and has strictly positive power at alternatives with $\mu_1 > 0$, while the zero test has zero power everywhere.

This example does not contradict d-admissibility. A d-dominating test would also have to satisfy

$$E_\mu[\psi(X)] \leq E_\mu[\phi(X)] = 0 \qquad \text{for all } \mu \in \Lambda_0,$$

which forces no increase in rejection probability on the null. Thus exact size is needed for the α-admissibility conclusion, not for the d-admissibility conclusion.

References

[1] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Third edition. Springer.